Logr: A habit logging system

Vityarthi project

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Programming in Java

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Introduction

Project Overview

The Student Habit Tracker is a simple Java-based application that helps students build and maintain positive habits through daily progress tracking and reporting.

Java Application

Console-based implementation

Progress Tracking

Daily habit completion monitoring

OOP Concepts

Classes, objects, and modular design

Problem Statement

Student Challenge

Students often struggle to maintain consistent daily habits due to lack of systematic tracking and accountability.

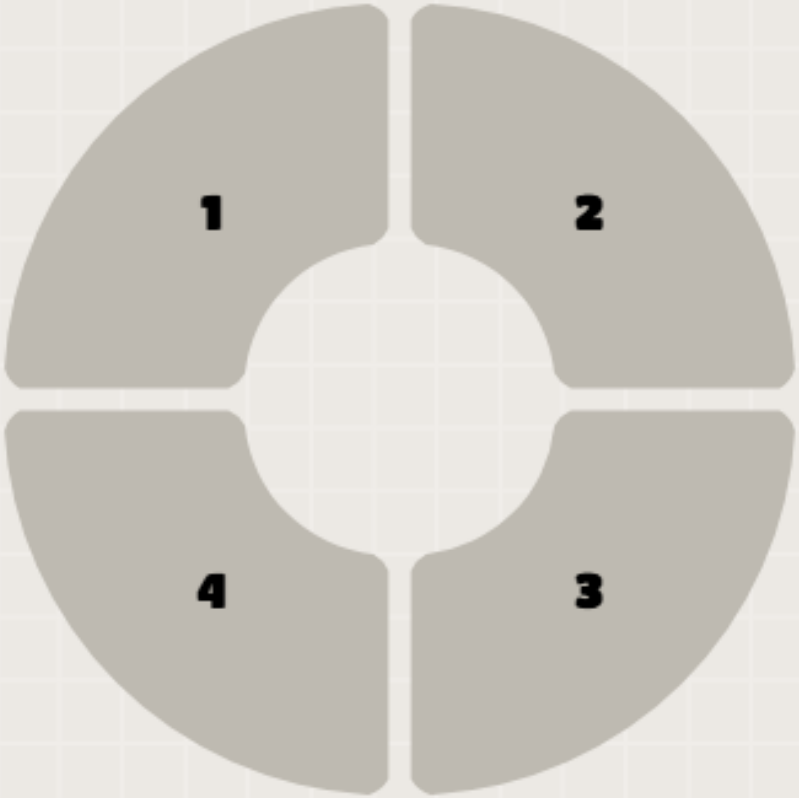
Solution Requirements

Need for a simple console-based application to add habits, mark daily progress, and generate reports.

Project Objectives

Develop Application

Create simple Java habit tracker



Habit Management

Add, delete, and view habits

Generate Reports

Summary and analytics

Progress Tracking

Mark daily completion

Functional Requirements

Habit Management

Add new habits, delete existing ones, and view complete habit list

Progress Tracking

Mark habits as completed for each day with date tracking

Report Generation

Display completion summary and total completed days for each habit

Non-Functional Requirements

Usability

Easy-to-use text-based menu interface

Performance

Instant response for all operations

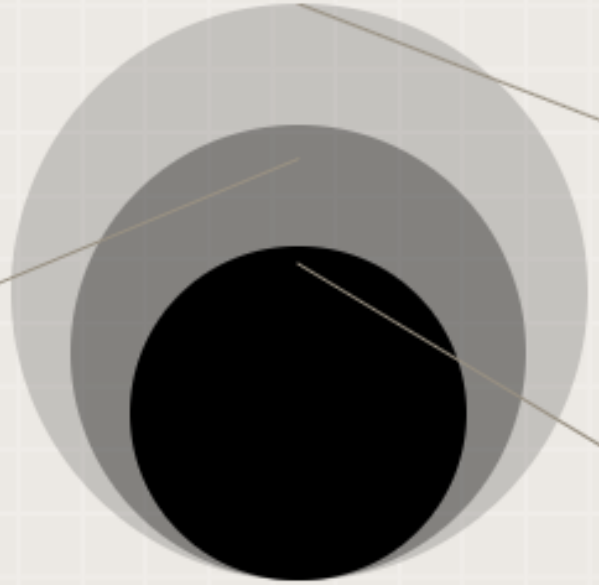
Scalability

Modular design for future enhancements

Reliability

Proper error handling and validation

System Architecture



Presentation Layer

HabitTrackerMain - user interface and menu

Service Layer

HabitService & ReportService - business logic

Data Layer

Habit class - stores habit data and properties

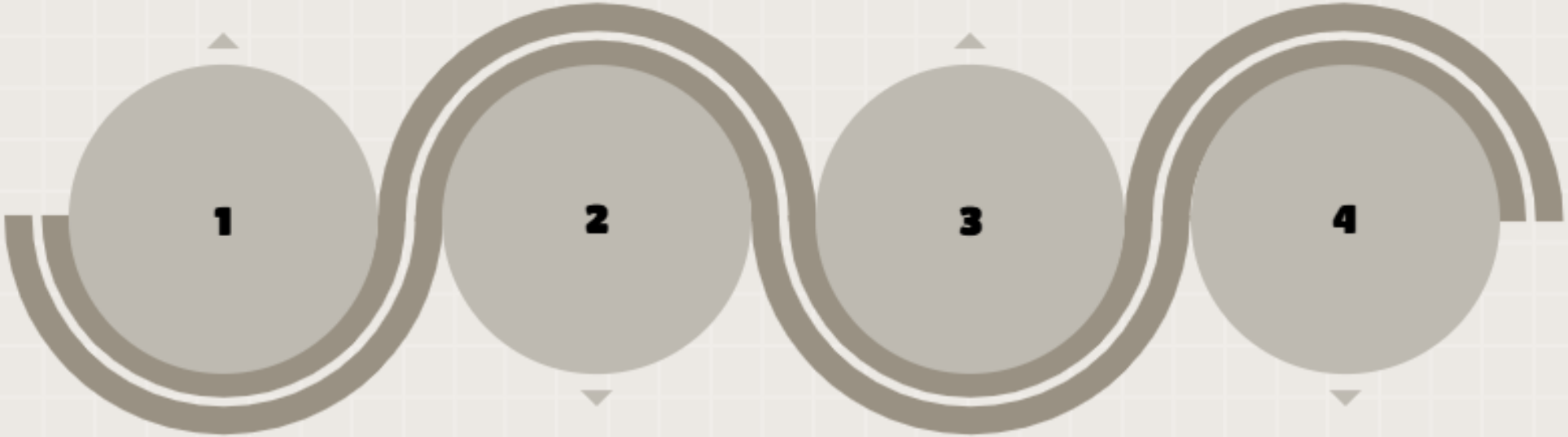
Workflow Process

Launch Program

User starts the Java application

Process Request

Service layer handles operations



Display Menu

System shows available options

Show Results

Output displayed on console

Use Case Diagram

1

Add Habit

Create new study or personal habits

2

Delete Habit

Remove unwanted habits from list

3

Track Progress

Mark daily completion status

4

View Reports

Generate habit completion summaries

Student User

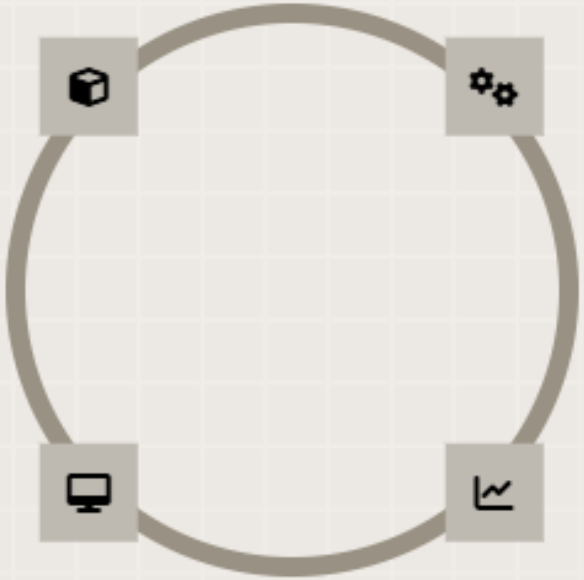
Class Diagram Structure

Habit Class

Core model class storing habit name, creation date, and completion status

HabitTrackerMain

Main class handling user interface and program flow



HabitService

Business logic for habit management operations

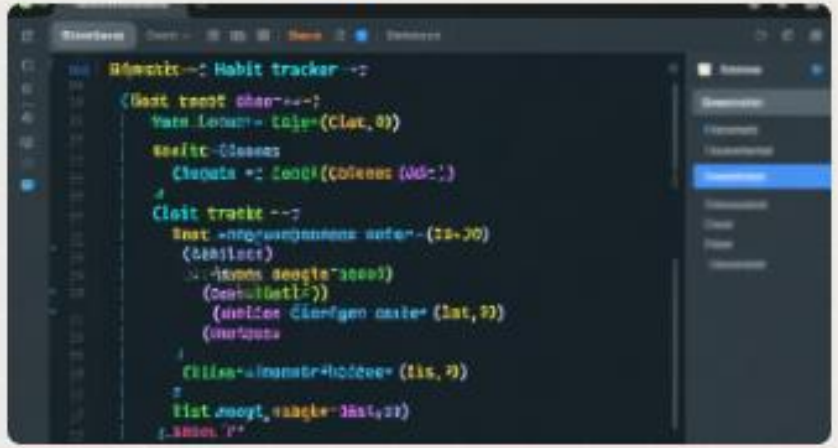
ReportService

Analytics and reporting functionality

Implementation Details

Technical Stack

Developed using Java JDK with ArrayList for data storage. Four modular classes created following OOP principles and standard coding practices.

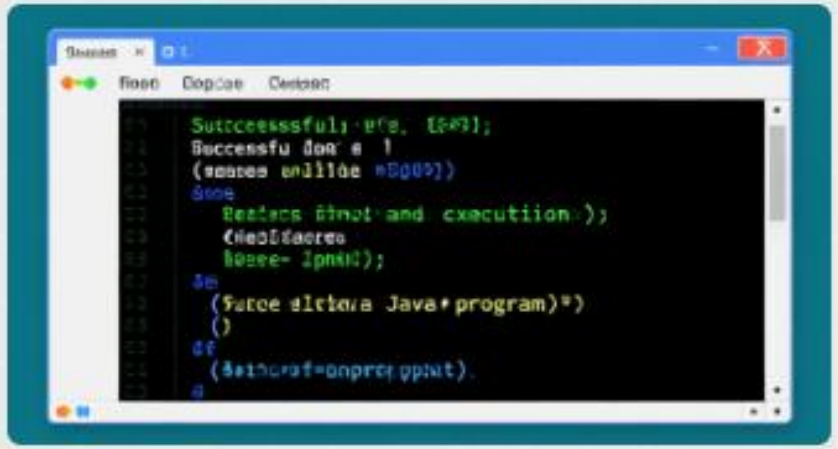


Java Classes

4 modular classes for clean architecture

Command Line

Compiled and executed through CMD



Sample Output Screenshots

Add Habit

User adding new study habits to track

View Habits

Displaying all active habits in the system

Mark Progress

Marking daily completion status

Testing Approach

Manual Testing

Comprehensive testing of each menu option and functionality

Edge Cases

Tested empty habit names, non-existent deletions, and empty list scenarios

Verification

Verified accurate counting of completed days and proper error handling

Challenges Faced

1

CMD Compilation

Learning to compile and run Java using command prompt

2

Input Validation

Handling invalid user input and edge cases

3

Modular Structure

Understanding clean code architecture principles

4

File Management

Proper naming and organization of Java source files

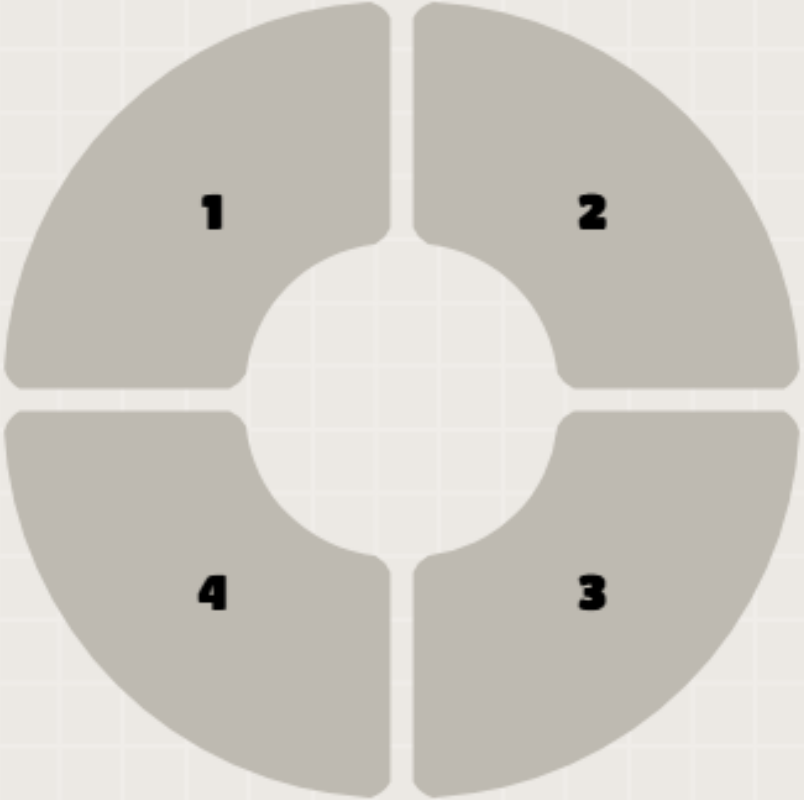
Key Learnings

OOP Concepts

Practical understanding of classes and objects

Data Structures

Java ArrayList and console input handling



Problem Solving

Debugging and systematic approach to coding

Development Process

Complete software project lifecycle experience